

Quicksort in TypeScript — sample capture

Source: ChatGPT · Captured: 12 Jun 2026, 09:29 pm · 4 messages · ~265 tokens

USER

Write a quicksort in TypeScript.

ASSISTANT

Here is a simple implementation:

```
ts:
function quicksort(arr: number[]): number[] {
  if (arr.length <= 1) return arr;
  const [p, ...rest] = arr;
  const less = rest.filter(x => x < p);
  const more = rest.filter(x => x >= p);
  return [...quicksort(less), p, ...quicksort(more)];
}
```

Note: this is `_not_` in-place.

USER

Make it in-place.

ASSISTANT

Sure:

```
ts:
function quicksortInPlace(arr: number[], lo = 0, hi = arr.length - 1): void {
  if (lo >= hi) return;
  const pivot = arr[hi];
  let i = lo;
  for (let j = lo; j < hi; j++) {
    if (arr[j] < pivot) {
      [arr[i], arr[j]] = [arr[j], arr[i]];
      i++;
    }
  }
  [arr[i], arr[hi]] = [arr[hi], arr[i]];
  quicksortInPlace(arr, lo, i - 1);
  quicksortInPlace(arr, i + 1, hi);
}
```